Logic strategy guide (version 1.0.0)

The ungraded lessons shown at the start of each guided-strategy backward branch, by branch ID. Each worked example is a different puzzle from every scored question.

S1-B1 · Reflect one cell at a time

Strategy: reflect one cell at a time

A mirror line acts like a fold. Each point keeps its distance from the line but ends up on the other side. Points on the line stay where they are.

For a vertical mirror line, a cell keeps its row and only its column changes. For a horizontal mirror line, a cell keeps its column and only its row changes.

So reflect one cell at a time: count how far it is from the line, then count the same distance on the other side. Check two or three cells; if any one is wrong, the figure is not the reflection.

Worked example (a different figure)

A grid has 6 rows. A horizontal mirror line lies between rows 3 and 4. A shape has cells at row 1 column 2, row 2 column 2 and row 3 column 5. Where is its reflection?

1. Row 3 is 1 row above the line, so its image is 1 row below it: row 4. Row 2 goes to row 5 and row 1 to row 6.
2. Columns do not change in a horizontal mirror.
3. Image: row 6 column 2, row 5 column 2 and row 4 column 5.
4. Quick check: each cell and its image add up to 7 rows (1 + 6, 2 + 5, 3 + 4), the same for every pair, because the line is halfway between them.

S2-B1 · Rewrite each statement in plain form

Strategy: rewrite each statement in plain form

Before solving, rewrite every statement so that it says one plain thing.

Double negatives cancel: "not untrue" means "true", and "did not fail to" means "did".

Units: decide whether a number counts items or groups (pairs, packs, dozens) and convert.

"At least 3" means 3, 4, 5, ...; "at most 3" means 0, 1, 2 or 3. List the values that are still possible.

A sentence about what someone SAID or THINKS is not a fact about the world.

Worked example (a different situation)

"It is not untrue that the library is not open." Is the library open?

Eggs come in cartons of a dozen. Pia needs 30 eggs. How many cartons must she buy?

1. "Not untrue" means "true", so the sentence says: the library is not open.
2. Cartons hold 12 eggs. 2 cartons give 24 eggs, which is too few; 3 cartons give 36, which is enough. She must buy 3 cartons.

S4-B1 · Cross out, one clue at a time

Strategy: cross out, one clue at a time

Write down every candidate first: all 12 months, all 7 days, every number in range.

Apply ONE clue at a time and cross out every candidate it rules out. Do not skip a candidate because it "seems" unlikely.

When the clues are done, exactly one candidate should be left. If more than one is left, the clues are not enough; if none is left, recheck the work.

Worked example (a different list)

A number from 1 to 20: it is a multiple of 3; it is odd; it is greater than 10.

1. Multiples of 3: 3, 6, 9, 12, 15, 18.
2. Odd: 3, 9, 15.
3. Greater than 10: 15.
4. Exactly one is left, so the number is 15.

S6-B1 · Write the key as a table

Strategy: write the key as a table

A code is a rule that turns each symbol into another symbol (or a number).

Write the key as a two-row table: plain symbols on top, coded symbols below. To encode, read top to bottom; to decode, find the coded symbol on the bottom row and read up.

When the key must be found from an example, use the example to fill in the table, then check the key on EVERY symbol of the example before using it.

Worked example (a different code)

A code replaces each digit d by 9 - d. What is 274 in code, and what number does the code 815 come from?

1. Table: 0->9, 1->8, 2->7, 3->6, 4->5, 5->4, 6->3, 7->2, 8->1, 9->0.
2. Encode 274: 2->7, 7->2, 4->5, so 725.
3. Decode 815: 8 comes from 1, 1 from 8, 5 from 4, so 184.
4. Check: 184 encodes to 815.

S7-B1 · Place the most constrained item first

Strategy: place the most constrained item first

Draw a row of blanks, one per position.

Fill in fixed positions first ("last", "2nd from the left").

Then join items that must be next to each other into a block ("right after").

Finally use before/after clues to order the remaining pieces, and check every clue at the end.

Worked example (a different line)

Four cars wait at a light: Pia's car is last. Quin's car is right in front of Rose's. Sal's car is in front of Quin's. Order the cars from front to back.

1. Pia is 4th.
2. Quin-Rose is a block of two.
3. Sal is in front of Quin, so the order of the front three is Sal, Quin, Rose.
4. Answer: Sal, Quin, Rose, Pia. Check each clue: all true.

S9-B1 · Follow one marked corner

Strategy: follow one marked corner

Pick one distinctive cell of the shape, for example the end of its longest arm, and note which way the shape bends from there (clockwise or counterclockwise).

A turn moves that cell around but keeps the bending direction the same.

A flip reverses the bending direction. If the direction is reversed, the figure is a mirror image and no turn can produce it (unless the shape is symmetric).

For a quarter-turn clockwise, the top of the shape goes to the right side; for a half-turn, the top goes to the bottom and left goes to right.

Worked example (a different shape)

An L made of 3 cells going down and 1 cell to the right at the bottom. Where does the single cell go after a half-turn?

1. After a half-turn, the bottom goes to the top and the right goes to the left.
2. So the single cell ends up at the TOP, on the LEFT: the shape looks like a "7" made of cells.
3. Bending direction: going down the long arm, the foot turns left (counterclockwise) in the original and still turns the same way after the half-turn, so it is a turn, not a flip.

S11-B1 · List the letters with their positions

Strategy: list the letters with their positions

Write the word in capitals with a position number under each letter: 1, 2, 3, ...

For "alphabetical order", compare each letter only with the one before it.

For counting, mark each match as you find it, then count the marks.

For first letters or acrostics, write the first letter of each word in order before reading the result.

Worked example (a different word)

Is GHOST in alphabetical order? How many times does O appear in COOKBOOK?

1. G(1) H(2) O(3) S(4) T(5): G <= H, H <= O, O <= S, S <= T. Yes, in order.
2. C O O K B O O K: O at positions 2, 3, 6, 7. O appears 4 times.

S14-B1 · Mark X for "no" and O for "yes"

Strategy: mark X for "no" and O for "yes" in a grid

Draw a grid: people down the side, options across the top.

For each "not" clue, put an X in that cell.

When a row has only one empty cell left, put an O there; then put X in the rest of that column (each option is used once).

Repeat until every row has an O. Then check every clue again.

Worked example (a different puzzle)

Three hikers, Uri, Val and Wen, carry a map, a rope or a stove (one each). Val does not carry the rope or the stove. Wen does not carry the rope.

1. Val: X rope, X stove, so O map. Put X under "map" for Uri and Wen.
2. Wen: X rope, X map, so O stove.
3. Uri gets the rope. Check: every clue holds.

S17-B1 · Suppose, then test

Strategy: suppose, then test

Pick one speaker and SUPPOSE that person tells the truth.

Then everything that person says is true. Use it to decide others, one step at a time. A liar's statement must be false.

If you reach a contradiction, the supposition was wrong: that speaker is a liar. Start again with that fact.

When every statement agrees with the speaker's type, you have a solution. Check that no other choice works if the question expects one answer.

Worked example (different people)

Pip says: "Rue and I are both liars." Rue says nothing. Who is what?

1. Suppose Pip is truthful. Then Pip is a liar: contradiction.
2. So Pip is a liar, and "Rue and I are both liars" is false.
3. Pip IS a liar, so for the statement to be false, Rue must be a truth-teller.
4. Answer: Pip liar, Rue truth-teller.

S20-B1 · The cut-off sheet is underneath

Strategy: the cut-off sheet is underneath

Look at one place where two sheets overlap.

The sheet whose outline runs straight through the overlap is on top there. The sheet whose outline stops at the other sheet's edge is underneath.

Record each pair as "X under Y". Then find the sheet that is under no other sheet: that is the top. Remove it and repeat.

For stacks of centered discs, a disc shows only if every disc above it is smaller.

Worked example (different sheets)

Three sheets: the outline of K stops at the edge of M; the outline of M stops at the edge of J; the outline of K stops at the edge of J.

1. Pairs: K under M, M under J, K under J.
2. J is under nothing: J is on top.
3. Then M, then K at the bottom. Order from bottom to top: K, M, J.

S23-B1 · Skip one square to find the opposite face

Strategy: skip one square to find the opposite face

A cube has 3 pairs of opposite faces. Opposite faces never share an edge.

In any straight row or column of the net, two squares with exactly one

square between them end up opposite.

Squares that share an edge in the net always share an edge on the cube.

For a square sticking out of a row, slide it along its row in your mind: it is opposite the square two steps away along the direction it folds.

Worked example (a different net)

A net has the row 1, 2, 3, 4 with square 5 above square 2 and square 6 below square 3.

1. In the row: 1 and 3 are opposite; 2 and 4 are opposite.
2. That leaves 5 and 6, which must be opposite each other.
3. Check: 5 touches 2 in the net, and 2 is not opposite 5. Every face has exactly one opposite.

S25-B1 · Number the positions

Strategy: number the positions

Write the row, and under it the position numbers 1, 2, 3, ... from the left.

Do one move at a time and rewrite the whole row after each move.

To find the fewest swaps of neighbors, count the pairs that are in the wrong order: each swap of neighbors fixes exactly one such pair.

Worked example (a different row)

Row: B, A, D, C. Fewest swaps of neighbors to reach A, B, C, D?

1. Pairs in the wrong order: (B, A) and (D, C). That is 2.
2. Swap B and A: A, B, D, C. Swap D and C: A, B, C, D. Two moves.

S28-B1 · Mark the route as you trace it

Strategy: mark the route as you trace it

Write each route as a list of towns, for example A-B-D.

To count routes, list them in a fixed order: all routes starting A-B first, then those starting A-C, and so on. Never revisit a town in the same route if the rule forbids it.

For the shortest route, find all towns 1 road away, then all towns 2 roads away, and so on.

For a walk that uses every road once, count roads at each town: it is possible only if 0 or 2 towns have an odd number of roads (and the network is all one piece).

Worked example (a different network)

Towns W, X, Y, Z with roads W-X, X-Y, Y-Z, W-Y. How many routes from W to Z without repeating a town?

1. Routes must end with Y-Z, because Z only touches Y.
2. From W to Y: W-Y, or W-X-Y.
3. So the routes are W-Y-Z and W-X-Y-Z: 2 routes.

S31-B1 · Draw the neighbor graph

Strategy: draw the neighbor graph

Put one dot for each region.

Walk around each region's border and draw a line to every region it shares a border with. A corner touch is not a border.

The number of lines at a dot is the number of neighbors.

To color the map, color the dots so that every line joins two different colors. Start with the region that has the most neighbors.

Worked example (a different map)

Four regions W, X, Y, Z: W borders X and Y; X borders Y and Z; Y borders Z.

1. Lines: W-X, W-Y, X-Y, X-Z, Y-Z.
2. W, X, Y all border each other, so they need 3 different colors: W red, X blue, Y green.
3. Z borders X and Y, so Z must be red. Three colors are enough.

S33-B1 · Change the letters that must change, one at a time

Strategy: change the letters that must change, one at a time

Write the start word above the end word and circle each position where they differ. Each step changes one letter, so you need at least as many steps as circled positions.

Try changing one circled letter at a time, and check that each new word is on the list.

If no circled letter can change to a listed word, you may need a detour: change a letter that is already correct and change it back later. Detours add steps.

Worked example (a different list)

List: COT, CAT, HAT, HOT, HUT. Ladder from COT to HAT?

1. COT and HAT differ in 2 positions (C/H, O/A): at least 2 steps.
2. COT -> CAT (O to A) -> HAT (C to H). Both words are listed, so 2 steps.

S34-B1 · Write each clue as a condition on numbers

Strategy: write each clue as a condition on numbers

Give each person a letter that stands for their position number.

Translate each clue: "somewhere left of" becomes <, "next to" becomes a

difference of 1 in either order, "exactly one between" becomes a difference of 2 in either order.

To test an arrangement, put its numbers into every condition. One failed condition rules the arrangement out.

Worked example (a different puzzle)

Three books, Map (M), Novel (N) and Poems (P), stand on a shelf in spots 1 to 3. Clues: Map is next to Poems; Novel is not in spot 2.

1. Conditions: |M - P| = 1 and N is 1 or 3.
2. If N = 1, then M and P fill 2 and 3, which are next to each other: N, M, P or N, P, M.
3. If N = 3, then M and P fill 1 and 2: M, P, N or P, M, N. Four arrangements fit.

S35-B1 · List every case, then read must, could and cannot

Strategy: list every case, then read must, could, cannot

Find every arrangement that meets all the conditions. Organize the list by one item with few choices, so no case is missed.

A statement MUST be true if it is true in every arrangement on the list.

It COULD be true if it is true in at least one. It CANNOT be true if it is true in none.

Worked example (a different puzzle)

Three chores, dishes (D), laundry (L) and sweeping (W), are done on days 1, 2, 3, one per day. Conditions: laundry is not on day 3; dishes are after sweeping.

1. Laundry on day 1: W, D fill days 2 and 3 with W first: L, W, D.
2. Laundry on day 2: W on day 1, D on day 3: W, L, D.

3. Dishes are on day 3 in both, so "dishes are on day 3" must be true. "Laundry is on day 1" could be true. "Sweeping is on day 3" cannot be true.

S36-B1 · Make an organized list

Strategy: make an organized list

Pick the choice with the fewest options and go through its values in order.

For each value, list what the rest can be, again in order.

When every value has been handled, the list is complete; count it.

Worked example (a different count)

How many two-digit numbers have digits that add up to 4?

1. Go through the tens digit in order: 1 gives 13, 2 gives 22, 3 gives 31, 4 gives 40.
2. A tens digit of 5 or more is too big, and 0 is not allowed for a two-digit number. So there are 4.

S38-B1 · Color the board like a checkerboard

Strategy: color the board like a checkerboard

Color the squares in a checkerboard pattern: squares that share a side get different colors.

Every domino covers one dark and one light square. So a region can be tiled only if it has equally many dark and light squares.

Equal counts do not guarantee a tiling. Then look for a square with only one neighbor: its domino is forced.

Worked example (a different region)

A T shape: three squares in a row, plus one square below the middle one. Can it be tiled by 2 dominoes?

1. Colors: the middle square is dark; the two ends and the square below are light. That is 1 dark and 3 light.
2. Each domino needs one dark square, so at most one domino fits. No tiling exists.

S40-B1 · Assume the opposite and look for the impossible

Strategy: assume the opposite and look for the impossible

Write the exact opposite (negation) of what you want to prove, and assume it.

Reason from that assumption with definitions and facts you know.

Stop when you reach something impossible, such as 1 = 0 or a number that is both even and odd. Then the assumption is false, so the statement is true.

Worked example (a different claim)

Prove: there is no smallest positive fraction.

1. Assume the opposite: there is a smallest positive fraction r.
2. Then r / 2 is also a positive fraction, and r / 2 < r.
3. That contradicts r being smallest, so no smallest positive fraction exists.

S41-B1 · Keep track of odd and even only

Strategy: keep track of odd and even only

Ask what happens to odd/even at each step. Does it switch every time, or stay the same?

If it switches every step, after an even number of steps it is back where it

began; after an odd number it is the other way.

A target with the wrong parity can never be reached, however the steps are chosen.

Worked example (a different puzzle)

A card shows its red side. Each turn it is flipped over. After 25 turns, which side shows?

1. Each flip switches the side. After an even number of flips it shows red again.
2. 25 is odd, so after 25 flips the other side shows.

S44-B1 · Look for something that never changes

Strategy: look for something that never changes

Write down how one move changes simple quantities: a total, a count, odd/even, or a remainder.

If some quantity is the same after every move, it is an invariant.

Compare the invariant at the start and at the target. If they differ, the target can never be reached.

Worked example (a different puzzle)

A counter shows 14. A move adds 6 or subtracts 9. Can it ever show 20?

1. Both moves change the count by a multiple of 3, so the remainder after dividing by 3 never changes.
2. 14 leaves remainder 2; 20 leaves remainder 2 too, so this test does not rule it out. Indeed 14 + 6 = 20.

S45-B1 · Fill the most constrained cell first

Strategy: fill the most constrained cell first

For each empty cell, list the letters its row and column have not used yet.

A cell with only one possible letter is forced: fill it in.

Also look along a row: if a letter can go in only one cell of that row, it goes there.

Repeat. Each new letter removes options from other cells.

Worked example (a different grid)

Rule: each row and column of a 3 by 3 grid holds X, Y and Z once. Row 1 is X Y Z. Row 2 starts with Z.

1. Row 2, column 2 cannot be Y (column) or Z (row), so it is X. Row 2 is Z X Y.
2. Row 3 gets the letter missing from each column: Y Z X.

S46-B1 · Find the gap between premise and conclusion

Find the gap between premise and conclusion

Write the premise and the conclusion side by side.

Find a word or idea in the conclusion that the premise never mentions, or a jump from "some" to "all" or from "then" to "now".

The assumption is the statement that bridges that gap. Test it: if the assumption were false, the conclusion should no longer follow.

Worked example (a different argument)

"Leo practiced piano every day this month, so he will win the recital prize."

1. Gap: daily practice to winning the prize.
2. Needed assumption: practicing every day is enough to play better than the other players. If it were false, the conclusion would not follow.

S47-B1 · List what each clue leaves

Strategy: list what each clue leaves

List every possibility allowed before any clue.

Cross out the ones a clue rules out. If exactly one answer remains, the clue is enough.

Do this for each clue alone, then for the clues together.

Worked example (a different question)

n is a whole number from 1 to 10. What is n? Clue (1): n is odd. Clue (2): n is a multiple of 3.

1. Clue (1) leaves 1, 3, 5, 7, 9: not enough. Clue (2) leaves 3, 6, 9: not enough.
2. Together: 3 and 9. Still not enough: a third clue is needed.

S49-B1 · Name what the argument does instead of giving reasons

Strategy: name what the argument does instead of giving reasons

Find the conclusion. Then find what is offered as the reason.

Ask: is the reason about the claim, or about something else (the person, how popular it is, what came first)?

Match what the argument does to a definition: attacking the person, false choice, hasty generalization, circular reasoning, after-so-because, slippery slope, appeal to popularity, straw man.

Worked example (a different argument)

"Everyone in my class uses this calculator, so it must give correct answers."

1. Conclusion: the calculator gives correct answers. Reason: everyone uses it.
2. The reason is about popularity, not about accuracy: appeal to popularity.

S51-B1 · Find a good answer, then prove nothing beats it

Strategy: find a good answer, then prove nothing beats it

Try several cases in an organized way and keep the best so far.

Look for a limit (a bound): a reason every answer is at most some value.

If your best example reaches the bound, it is the best possible.

Worked example (a different problem)

Two whole numbers add to 9. What is the largest product?

1. Try 1 x 8 = 8, 2 x 7 = 14, 3 x 6 = 18, 4 x 5 = 20.
2. Moving away from the middle makes the product smaller, so 20 is the largest.

S52-B1 · Write the differences

Strategy: write the differences

Under the sequence, write the difference between each pair of neighbors.

Equal differences: rule (1). Differences that grow by the same amount: rule (3).

If neither, divide each term by the one before; equal results: rule (2).

Worked example (a different sequence)

Sequence: 4, 5, 8, 13, ...

1. Differences: 1, 3, 5. They grow by 2 each time: rule (3).

2. Next difference 7, so the next term is 13 + 7 = 20.

S53-B1 · Sort the letters

Strategy: sort the letters

Write the letters of each word in alphabetical order, keeping repeats.

Two words are anagrams exactly when their sorted letters match.

To check whether a word can be spelled from a set of letters, cross off each letter as you use it.

Worked example (different words)

Are DIRTY and DIRT anagrams? Are CAFE and FACE?

1. DIRTY sorts to D, I, R, T, Y and DIRT sorts to D, I, R, T. DIRTY has an extra Y, so they are not anagrams.
2. CAFE and FACE both sort to A, C, E, F: anagrams.

S54-B1 · Each weighing has three outcomes

Strategy: each weighing has three outcomes

A weighing can tip left, tip right or balance: three outcomes.

Split the suspects into three groups as equal as possible: two on the scale (same size) and one off it.

After w weighings there are at most 3 x 3 x ... (w times) outcome patterns, so at most that many coins can be told apart.

Worked example (a different count)

There are 3 coins and one is heavier. How many weighings?

1. Weigh coin 1 against coin 2. If one goes down, it is heavy. If they balance, coin 3 is heavy.

2. One weighing is enough.